Kitaev QAOA Experiment Pack

Scaling QAOA Ground-State Preparation for the Kitaev Chain to N = 8, 10, and 12 Sites on Noisy Superconducting Hardware

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Executive Summary

This experiment pack provides a complete, self-contained specification for preparing the ground state of the Kitaev chain Hamiltonian using the Quantum Approximate Optimization Algorithm (QAOA) on noisy superconducting quantum hardware. Three system sizes are targeted: N = 8, 10, and 12 sites, covering the deep topological phase (mu = 0, w = 1, Delta = 1). The pack encompasses the full pipeline from Hamiltonian-to-qubit mapping via the Jordan-Wigner transformation through validated runnable Qiskit and PennyLane circuit code, Particle Swarm Optimization (PSO) with adaptive shot scheduling, hardware noise characterization and mitigation, hardware-aware transpilation, resource estimation, a validation suite with exact diagonalization references, and a structured six-week runbook with milestones and fallback strategies.

At the deep topological point, the Kitaev chain hosts zero-energy Majorana edge modes — a paradigmatic example of symmetry-protected topological order. Faithful preparation of the ground state on hardware requires careful attention to fermion-parity conservation, decoherence budgeting, and error mitigation. This document provides actionable guidance on each of these challenges.

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Section 1: Hamiltonian-to-Qubit Mapping

1.1 The Kitaev Chain Hamiltonian

The Kitaev chain model (Kitaev, 2001) describes a one-dimensional wire of spinless fermions with p-wave superconducting pairing under open boundary conditions (OBC). It is the canonical model for realizing Majorana zero modes at chain endpoints. The second-quantized Hamiltonian is:

```
H =

–

mu

.

∑

j

(2n

j

–

1)

–

∑

j

[w

.
```

```
(c

j

†

c

j+1

+ h.c.) + Delta

.

(c

j

c

j+1

+ h.c.)]
```

Parameter definitions:

- **mu** — chemical potential; the topological phase requires |mu| < 2|w|.
- **w** — nearest-neighbor hopping amplitude (real, positive by convention).
- **Delta** — p-wave superconducting pairing gap (real, positive).
- $\mathbf{n_j = c_j^\dagger c_j}$ — fermion number operator at site j.
- $\mathbf{c_j}$, $\mathbf{c_j^\dagger}$ — fermionic annihilation and creation operators satisfying canonical anticommutation relations.
- **h.c.** — Hermitian conjugate.

The model has three distinct phases: a *topological* superconducting phase for |mu| < 2|w| (hosting zero-energy Majorana edge modes), and a *trivial* phase for |mu| > 2|w|. The phase boundary at |mu| = 2|w| is a quantum critical point described by

the Ising universality class. The special **deep topological point**, used as the default throughout this pack, is:

```
mu = 0,

w = 1,

Delta = 1
```

At this point the model is exactly solvable via Bogoliubov transformation, the gap is maximized at Delta_gap = 2w, and the Majorana edge modes are perfectly localized at the endpoints (zero correlation length). This makes it an ideal benchmark target for a QAOA ground-state preparation experiment.

1.2 Jordan-Wigner Transformation

The Jordan-Wigner (JW) transformation maps fermionic operators to qubit (Pauli) operators while preserving the fermionic anticommutation algebra via a nonlocal string of Z operators. The mapping for site j in an N-site chain is:

```
c

j

= (

∏

k < j

Z

k

)

⊗
```

(X

j

+ i

.

Y

j

) / 2

c

j

†

= (

∏

k < j

Z

k

)

⊗

(X

j

−

i

.

Y

j

) / 2

The relevant bilinear combinations that appear in the Hamiltonian transform as follows:

Fermionic Term	Qubit Operator (JW Image)	Type
$n_j = c_j^\dagger c_j$	$(I - Z_j) / 2$	Number operator
$c_j^\dagger c_{j+1} + \text{h.c.}$	$(1/2)(X_jX_{j+1} + Y_jY_{j+1})$	Hopping (XX+YY)
$c_j c_{j+1} + \text{h.c.}$	$(1/2)(X_jY_{j+1} - Y_jX_{j+1})$	Pairing (XY−YX)
$2n_j - 1$	$-Z_j$	Chemical potential

A crucial property of the JW transformation for the Kitaev chain is that all resulting Pauli terms are *strictly nearest-neighbor* — no long-range Pauli strings appear in the qubit Hamiltonian. This is because the JW string cancels exactly between adjacent-site bilinears, making the mapped Hamiltonian directly compatible with a linear qubit topology (heavy-hex or ladder connectivity) without any SWAP overhead for N up to approximately 12 sites.

1.3 Resulting Qubit Hamiltonian

Applying the JW transformation to the full Kitaev chain Hamiltonian and collecting terms yields the qubit Hamiltonian:

H

qubit

=

∑

j

[mu/2

.

Z

j

]

−

∑

j=0

N−2

[w/2

.

(X

j

X

j+1

+ Y

j

Y

j+1

)]

−

∑

j=0

N−2

[Delta/2

.

(X

j

Y

j+1

−

Y

j

X

j+1

)]

The first sum runs over all N sites; the second and third sums run over N−1 bonds under OBC. The constant energy offset from the identity operator has been dropped. The Pauli term inventory for the three system sizes is:

N	Qubits	Z terms	XX+YY terms	XY−YX terms	Total Pauli terms
8	8	8	14	14	36
10	10	10	18	18	46
12	12	12	22	22	56

Note that XX+YY and XY−YX each contribute 2(N−1) terms split into individual Pauli products (e.g., XX counted separately from YY). Total Pauli terms scale linearly as 4N−4, making the Hamiltonian sparse and amenable to efficient circuit implementation.

1.4 Parity Symmetry

The JW-mapped Hamiltonian commutes with the total fermion parity operator:

$$P = \prod_{j=0}^{N-1} Z_j = Z_0 \otimes Z_1 \otimes$$

...

⊗

Z

N−1

$[H_{qubit}, P] = 0$ for all values of mu, w, and Delta. Consequently, the Hilbert space decomposes into two parity sectors: even parity (P eigenvalue +1, even number of |1⟩ qubits) and odd parity (−1, odd number of |1⟩ qubits).

At the deep topological point mu = 0, w = 1, Delta = 1, the ground state lies in the **even parity sector (P = +1)**. This has two important practical consequences:

- **Initial state design:** The QAOA initial state must have even parity. The half-filling state |1010...10⟩ satisfies this for all even N.
- **Parity postselection:** Measurement shots with an odd number of 1-bits are discarded. In the ideal case exactly 50% of shots survive; on noisy hardware, typical acceptance fractions are 40–60%. Fractions below 20% indicate a mixer or transpilation error.

Tip: Enforcing Parity Conservation

Use the XY (iSWAP-type) mixer $H_{mix} = (1/2)\sum_j(X_jX_{j+1} + Y_jY_{j+1})$, which conserves total Z (= fermion number mod 2 = parity). This keeps the QAOA state in the even parity sector throughout optimization without any ancilla or measurement feedback.

1.5 Exact Diagonalization Reference Values

The following reference energies are obtained by constructing the full $2^N \times 2^N$ qubit Hamiltonian matrix and diagonalizing via dense LAPACK routines

(numpy.linalg.eigh). These values serve as ground truth for all simulator and hardware validation steps.

N	E_{exact}	E_{exact} / N	Spectral gap E_{gap}	Hilbert space dim.
8	−5.656854	−0.707107	1.172480	256
10	−7.071068	−0.707107	0.938069	1,024
12	−8.485281	−0.707107	0.783460	4,096

Note that $E_{exact}/N = -1/\sqrt{2} \approx -0.707107$ exactly for all N at the deep topological point (mu = 0, w = Delta = 1). The spectral gap closes as $N \to \infty$ (bulk gap = 2w = 2 in the thermodynamic limit; the finite-size gap converges to this from below). Phase boundaries are located at mu = ±2 (for w = 1): topological for −2 < mu < +2; trivial for |mu| > 2.

Section 2: QAOA Circuit Templates

2.1 QAOA Ansatz Structure

The QAOA ansatz of depth p is the product-formula state:

```
|
ψ
p
(
γ
,
```

β

)

⟩

=

∏

k=1

p

[e

−

i

β

k

H

mix

.

e

−

i

γ

k

H

cost

] |

ψ

0

⟩

where the parameter vectors are $\gamma = (\gamma_1, \ldots, \gamma_p)$ (cost angles) and $\beta = (\beta_1, \ldots, \beta_p)$ (mixer angles), giving 2p free parameters total. The initial state is:

|

ψ

0

⟩

= |1010...10

⟩

(half-filling, even parity, N/2 excitations)

This Neel-like state lies at half-filling and has even parity for all even N, making it a natural starting point. It is also a reasonable approximation to the ground state structure in the large-mu regime and connects adiabatically to the topological ground state as parameters are optimized.

The cost Hamiltonian $H_{cost} = H_{qubit}$ (the full JW-mapped Kitaev Hamiltonian). The mixer H_{mix} is the parity-preserving XY mixer defined in Section 2.3.

2.2 Cost Layer Decomposition

Each term in the cost Hamiltonian exponential $\exp(-i\theta\, P_j P_{j+1})$ is a two-qubit Pauli rotation. The standard decomposition into native gates (CX, H, S, Rz) is:

Rotation	Gate Sequence	CX count
exp(−iθ XX)	H – CX – Rz(2θ) – CX – H	2
exp(−iθ YY)	$S^{\dagger}$–H – CX – Rz(2θ) – CX – H–S	2
exp(−iθ XY)	H–$S^{\dagger}$–H – CX – Rz(2θ) – CX – H–H–S	2
exp(−iθ YX)	$S^{\dagger}$–H–H – CX – Rz(2θ) – CX – H–S–H	2

For each bond j in the cost layer, four such rotations are applied sequentially (XX, YY, XY, YX), contributing 8 CX gates per bond. Over N−1 bonds per cost layer at depth p, the total CX count from the cost layers is 8(N−1)p. Adjacent XX and YY rotations on the same bond can be combined into a single RXX+RYY block that shares CX gates, reducing the cost to 4 CX per bond per term pair in optimized transpilation.

2.3 Parity-Preserving XY Mixer

The XY mixer Hamiltonian is:

H

mix

= (1/2)

∑

j=0

N−2

(X

j

X

```
j+1

+ Y

j

Y

j+1

)
```

This operator conserves the total magnetization $\sum_j Z_j$ and therefore conserves fermion parity $P = \prod_j Z_j$ (mod 2 conservation). The XY mixer implements iSWAP-type exchange interactions that swap $|01\rangle \leftrightarrow |10\rangle$ with a phase, keeping the total excitation number fixed. This is essential for staying in the correct parity sector throughout the QAOA optimization trajectory.

Staggered Trotter implementation: The mixer layer $\exp(-i\beta H_{mix})$ is implemented by applying bonds in two layers:

- **Even bonds:** apply $\exp(-i\beta (X_jX_{j+1} + Y_jY_{j+1}))$ for j = 0, 2, 4, ... (parallelizable)
- **Odd bonds:** apply the same for j = 1, 3, 5, ... (parallelizable)

Each bond rotation $\exp(-i\beta (XX+YY))$ is implemented as RXX(β) followed by RYY(β), at a cost of 2 CX gates per bond. The staggered structure avoids gate conflicts on adjacent qubits.

2.4 Qiskit Code — Full QAOA Circuit

The following parametrized Qiskit circuit builder generates the full QAOA ansatz for the Kitaev chain. It uses `ParameterVector` objects for γ and β, enabling parameter binding at evaluation time without circuit recompilation.

```
from qiskit import QuantumCircuit, QuantumRegister, ClassicalRegister from
qiskit.circuit import ParameterVector import numpy as np  def
build_kitaev_qaoa(N: int, p: int, mu=0.0, w=1.0, delta=1.0):     """
Build a parametrized QAOA circuit for the Kitaev chain.     N in {8, 10, 12};
p = QAOA depth (number of cost+mixer layers).     Returns a QuantumCircuit
with 2*p free parameters.     """     gamma = ParameterVector('\u03b3', p)
# cost angles     beta  = ParameterVector('\u03b2', p)   # mixer angles
qr = QuantumRegister(N, 'q')     cr = ClassicalRegister(N, 'c')     qc =
```

```
QuantumCircuit(qr, cr)      # --- Initial state: |1010...10> (half-filling,
even parity) ---     for j in range(0, N, 2):         qc.x(j)
qc.barrier(label='init')      for k in range(p):         # ---- Cost layer:
exp(-i * gamma[k] * H_cost) ----         # Z terms: exp(-i * gamma[k] *
(mu/2) * Z_j) = Rz(mu*gamma[k], j)         for j in range(N):             if
mu != 0.0:                qc.rz(mu * gamma[k], j)          # XX terms: exp(-
i * gamma[k] * (-w/2) * X_j X_{j+1})         # YY terms: exp(-i * gamma[k] *
(-w/2) * Y_j Y_{j+1})         # XY terms: exp(-i * gamma[k] * (-delta/2) *
X_j Y_{j+1})         # YX terms: exp(-i * gamma[k] * (+delta/2) * Y_j
X_{j+1})         for j in range(N - 1):             theta_hop   = -w     *
gamma[k]   # hopping angle             theta_pair  = -delta * gamma[k]   #
pairing angle             # XX rotation: H-CX-Rz(2*theta)-CX-H
qc.h(j); qc.h(j + 1)             qc.cx(j, j + 1)             qc.rz(2 *
theta_hop, j + 1)             qc.cx(j, j + 1)             qc.h(j); qc.h(j +
1)             # YY rotation: Sdg-H - CX - Rz - CX - H-S
qc.sdg(j); qc.h(j); qc.sdg(j + 1); qc.h(j + 1)             qc.cx(j, j + 1)
qc.rz(2 * theta_hop, j + 1)             qc.cx(j, j + 1)             qc.h(j);
qc.s(j); qc.h(j + 1); qc.s(j + 1)             # XY rotation: H-Sdg-H / H-H-S
qc.h(j); qc.sdg(j + 1); qc.h(j + 1)             qc.cx(j, j + 1)
qc.rz(2 * theta_pair, j + 1)             qc.cx(j, j + 1)             qc.h(j);
qc.h(j + 1); qc.s(j + 1)             # YX rotation: Sdg-H-H / H-S-H
qc.sdg(j); qc.h(j); qc.h(j + 1)             qc.cx(j, j + 1)
qc.rz(-2 * theta_pair, j + 1)   # note sign flip for YX vs XY
qc.cx(j, j + 1)             qc.h(j); qc.s(j); qc.h(j); qc.h(j + 1)
qc.barrier(label=f'cost_{k}')          # ---- Mixer layer: exp(-i * beta[k] *
H_mix) XY parity-preserving ----         # Even bonds first, then odd bonds
(staggered Trotter)         for j in range(0, N - 1, 2):       # even bonds
qc.rxx(beta[k], j, j + 1)             qc.ryy(beta[k], j, j + 1)         for j
in range(1, N - 1, 2):       # odd bonds             qc.rxx(beta[k], j, j +
1)             qc.ryy(beta[k], j, j + 1)
qc.barrier(label=f'mix_{k}')      qc.measure(qr, cr)     return qc   # ----
Example usage: N=8, p=2 ---- qc = build_kitaev_qaoa(N=8, p=2) print(f"Free
parameters : {qc.num_parameters}")   # 4  (2 gamma + 2 beta) print(f"Qubits
: {qc.num_qubits}")        # 8 print(f"Circuit depth   : {qc.depth()}")  #
Bind parameters and get statevector (noiseless) from qiskit.quantum_info
import Statevector import numpy as np bound =
qc.remove_final_measurements(inplace=False) params_example =
{qc.parameters[i]: np.pi/4 for i in range(qc.num_parameters)} sv =
Statevector(bound.bind_parameters(params_example)) print(f"Statevector norm:
{sv.probabilities().sum():.6f}")   # 1.000000
```

2.5 PennyLane Code — QNode Factory

The PennyLane implementation uses native `IsingXX`, `IsingYY`, and `PauliRot` operations for the cost layer, with parameter-shift-compatible differentiation. The `make_qaoa_qnode` factory returns both the compiled QNode and the Hamiltonian object for expectation-value evaluation.

```
import pennylane as qml import numpy as np  def make_qaoa_qnode(N, p, mu=0.0,
w=1.0, delta=1.0, shots=None):     """     PennyLane QAOA QNode factory for
the Kitaev chain.     Returns (circuit_fn, H_cost) where circuit_fn(params) -
> <H_cost>.     params: numpy array of shape (2*p,) — first p entries are
gamma, next p are beta.     """     # ---- Build the qubit Hamiltonian ----
coeffs, ops = [], []      # Z terms (chemical potential)     for j in
range(N):         coeffs.append(mu / 2.0)         ops.append(qml.PauliZ(j))
# Hopping and pairing terms     for j in range(N - 1):         coeffs += [-
```

```
w / 2.0,      # XX                    -w / 2.0,      # YY
-delta / 2.0,  # XY                     delta / 2.0]  # YX (note sign)
ops    += [qml.PauliX(j) @ qml.PauliX(j + 1),
qml.PauliY(j) @ qml.PauliY(j + 1),                    qml.PauliX(j) @
qml.PauliY(j + 1),                    qml.PauliY(j) @ qml.PauliX(j + 1)]
H = qml.Hamiltonian(coeffs, ops)      # ---- Device ----     dev =
qml.device('default.qubit', wires=N, shots=shots)      @qml.qnode(dev,
diff_method='parameter-shift')     def circuit(params):         gamma =
params[:p]         beta  = params[p:]          # Initial state |1010...10>
for j in range(0, N, 2):             qml.PauliX(wires=j)          # QAOA
layers         for k in range(p):             # -- Cost layer --
# Z terms             for j in range(N):                 if mu != 0.0:
qml.RZ(phi=mu * gamma[k], wires=j)              # Hopping: XX + YY  (coeff =
-w/2, rot angle = 2*coeff*gamma)            for j in range(N - 1):
qml.IsingXX(phi=2.0 * (-w / 2.0) * gamma[k], wires=[j, j + 1])
qml.IsingYY(phi=2.0 * (-w / 2.0) * gamma[k], wires=[j, j + 1])              #
Pairing: XY - YX  (using PauliRot)             for j in range(N - 1):
qml.PauliRot(theta=2.0 * (-delta / 2.0) * gamma[k],
pauli_word='XY', wires=[j, j + 1])                qml.PauliRot(theta=2.0 * (
delta / 2.0) * gamma[k],                           pauli_word='YX',
wires=[j, j + 1])             # -- XY mixer: parity-preserving --
for j in list(range(0, N - 1, 2)) + list(range(1, N - 1, 2)):
qml.IsingXX(phi=beta[k], wires=[j, j + 1])
qml.IsingYY(phi=beta[k], wires=[j, j + 1])          return qml.expval(H)
return circuit, H   # ---- Example usage: N=8, p=2 ---- circuit, H =
make_qaoa_qnode(N=8, p=2, mu=0.0, w=1.0, delta=1.0) params0 =
np.random.default_rng(42).uniform(0, np.pi, 4) energy  = circuit(params0)
print(f"Energy at random params : {energy:.4f}")  # Gradient via parameter-
shift (compatible with scipy optimizers) grad_fn = qml.grad(circuit) grads
= grad_fn(params0) print(f"Gradient vector         : {np.round(grads, 4)}")
```

Section 3: PSO Optimizer

3.1 Why PSO for QAOA

Particle Swarm Optimization (PSO) is the recommended classical optimizer for this experiment for several interconnected reasons. Standard gradient-based optimizers (ADAM, L-BFGS) require reliable gradient estimates, which become prohibitively expensive under shot noise for $p \geq 2$. Finite-difference and parameter-shift gradients require $O(p)$ additional circuit evaluations per step. Moreover, QAOA energy landscapes become increasingly non-convex with depth p, featuring exponentially many local minima — a setting where gradient descent frequently stagnates.

Recent benchmarking work (Bhat, Wang & Parampalli, arXiv:2506.06790v3, 2025) demonstrates that PSO consistently outperforms COBYLA and SPSA under shot-

noise conditions for QAOA optimization, with lower final energies and fewer circuit evaluations to convergence. Key PSO advantages for this application:

- **Gradient-free:** No additional circuit evaluations for gradient estimation; each particle requires exactly one energy evaluation per iteration.
- **Population-based:** Maintains a swarm of candidate solutions, enabling simultaneous exploration of multiple basins of attraction — critical for avoiding local minima in the QAOA landscape.
- **Shot-noise robust:** The collective global-best tracking is intrinsically smoothed by the population, reducing sensitivity to individual noisy function evaluations.
- **Adaptive inertia:** The decreasing inertia weight w(t) naturally transitions from exploration (early iterations) to exploitation (late iterations).
- **Warm-start compatible:** The INTERP strategy (Section 3.6) provides excellent initialization when scaling from p to p+1 layers.

3.2 PSO Algorithm

The standard PSO velocity and position update equations are:

v

i

(t+1) = w(t)

.

v

i

(t) + c

1

.

r

1

.

(pbest

i

–

x

i

(t)) + c

2

.

r

2

.

(gbest

–

x

i

(t))

x

i

(t+1) = x

i

(t) + v

i

(t+1)

where $x_i(t)$ is the position of particle i at iteration t, $v_i(t)$ is its velocity, $pbest_i$ is its personal best position, gbest is the global best position, and r_1, $r_2 \sim \text{Uniform}(0,1)$ are independent random vectors drawn fresh each iteration. The adaptive inertia weight follows a linear decay:

w(t) = w

start

–

(w

start

–

w

end

)

.

t / T

max

Positions are clipped to the parameter bounds after each update. Particles escaping the bounds are reflected inward (velocity reversal) to maintain diversity near boundaries.

3.3 Recommended Hyperparameters

N	p	Swarm size	Max iters	w_{start}	w_{end}	c_1	c_2	γ bounds	β bounds
8	1	20	80	0.9	0.4	2.0	2.0	[0, π]	[0, π/2]
8	2	25	100	0.9	0.4	2.0	2.0	[0, π]	[0, π/2]
10	2	30	120	0.9	0.4	2.0	2.0	[0, π]	[0, π/2]
10	3	35	140	0.9	0.4	2.0	2.0	[0, π]	[0, π/2]
12	3	40	150	0.9	0.4	2.0	2.0	[0, π]	[0, π/2]
12	4	45	180	0.9	0.4	2.0	2.0	[0, π]	[0, π/2]

3.4 Adaptive Shot Scheduling

To minimize total shot cost while maintaining convergence quality, a three-phase adaptive shot schedule is used. The rationale is that early PSO iterations require only coarse energy estimates to identify promising regions, while late iterations need high accuracy to distinguish nearby local minima and refine the global optimum.

Phase	Iteration range	Shots per eval	Purpose
Phase 1	0 – 30% of T_{max}	256	Rough landscape survey; identify low-energy regions
Phase 2	30 – 70% of T_{max}	1,024	Moderate

Phase	Iteration range	Shots per eval	Purpose
			accuracy convergence toward gbest basin
Phase 3	70 – 100% of T_{max}	4,096	High-accuracy refinement of gbest
Local refinement	Post-PSO (Nelder-Mead)	4,096	Polish gbest with gradient-free local search

3.5 Total Shot Budget

N	p	PSO shots	Local shots	Total shots	Circuit calls
8	1	1,280,000	200,000	1,480,000	~1,700
8	2	2,500,000	200,000	2,700,000	~2,700
10	2	3,600,000	200,000	3,800,000	~3,720
10	3	5,600,000	200,000	5,800,000	~4,940
12	3	7,200,000	200,000	7,400,000	~6,200
12	4	10,800,000	200,000	11,000,000	~8,345

3.6 PSO Implementation Code

```
# pso_optimizer.py — Full PSO with adaptive shots, restarts, and INTERP warm-start import numpy as np from scipy.optimize import minimize  class KitaevPSO:     """     Particle Swarm Optimizer for QAOA Kitaev chain.     Parameter layout: [gamma_1,...,gamma_p, beta_1,...,beta_p].     gamma bounds: [0, pi]; beta bounds: [0, pi/2].     """     def __init__(self, N: int, p: int, n_particles: int = 30,                  max_iters: int = 120, w_start: float = 0.9,                  w_end: float = 0.4, c1: float = 2.0, c2: float = 2.0,                  seed: int = 42):         self.N, self.p   = N, p         self.n_dim       = 2 * p          # total parameter count         self.n_particles = n_particles         self.max_iters   = max_iters         self.w_start, self.w_end = w_start, w_end         self.c1, self.c2 = c1, c2         self.rng = np.random.default_rng(seed)      def _shot_schedule(self, t: int) -> int:         frac = t / self.max_iters         if frac < 0.30: return 256         if frac < 0.70: return 1024         return 4096      def run(self, objective_fn):         """         objective_fn(params: np.ndarray, shots: int) -> float  (energy to minimise).         Returns (best_params, best_energy).         """         rng  = self.rng         n, d = self.n_particles, self.n_dim          # Parameter bounds         lo =
```

```
np.zeros(d)         hi = np.array([np.pi] * (d // 2) + [np.pi / 2] * (d //
2))          # Initialise swarm         x  = rng.uniform(lo, hi, (n, d))
v  = rng.uniform(-0.05 * (hi - lo), 0.05 * (hi - lo), (n, d))          #
Personal bests         pbest     = x.copy()         pbest_val = np.full(n,
np.inf)         for i in range(n):             pbest_val[i] =
objective_fn(x[i], shots=256)          gbest_idx = np.argmin(pbest_val)
gbest     = pbest[gbest_idx].copy()         gbest_val = pbest_val[gbest_idx]
no_improve = 0   # stagnation counter            for t in
range(self.max_iters):             w_t   = self.w_start - (self.w_start -
self.w_end) * t / self.max_iters              shots = self._shot_schedule(t)
r1    = rng.random((n, d))             r2    = rng.random((n, d))
# Velocity and position update             v = (w_t * v                  +
self.c1 * r1 * (pbest - x)                  + self.c2 * r2 * (gbest - x))
x = x + v             # Enforce bounds (reflect at boundaries)
lo_viol = x < lo; hi_viol = x > hi             x = np.clip(x, lo, hi)
v[lo_viol | hi_viol] *= -1.0   # reflect velocity             prev_gbest_val
= gbest_val              for i in range(n):                 val =
objective_fn(x[i], shots=shots)                 if val < pbest_val[i]:
pbest_val[i] = val                     pbest[i]     = x[i].copy()
if val < gbest_val:                         gbest_val = val
gbest     = x[i].copy()             # Stagnation detection             if
abs(gbest_val - prev_gbest_val) < 1e-6:                 no_improve += 1
else:                 no_improve = 0             # Restart scattered
particles if stagnating for 20 iterations             if no_improve >= 20:
scatter_idx = rng.choice(n, size=n // 3, replace=False)
x[scatter_idx] = rng.uniform(lo, hi, (len(scatter_idx), d))
v[scatter_idx] = rng.uniform(-0.05*(hi-lo), 0.05*(hi-lo),
(len(scatter_idx), d))                 no_improve = 0          # ---- Local
refinement with Nelder-Mead ----         res = minimize(lambda p_:
objective_fn(p_, shots=4096),                         gbest, method='Nelder-
Mead',                      options={'maxiter': 500, 'xatol': 1e-5,
'fatol': 1e-5})         if res.fun < gbest_val:             gbest, gbest_val
= res.x, res.fun          return gbest, gbest_val   def
interp_warmstart(params_p: np.ndarray) -> np.ndarray:     """     INTERP
warm-start: interpolate optimal p-layer parameters to p+1 layers.     Uses
linear interpolation along the depth axis.     Reference: Zhou et al. (2020),
arXiv:1812.01041.     """     p = len(params_p) // 2     gamma_p, beta_p =
params_p[:p], params_p[p:]     t_old = np.linspace(0, 1, p) if p > 1 else
np.array([0.5])     t_new = np.linspace(0, 1, p + 1)     gamma_new =
np.interp(t_new, t_old, gamma_p)     beta_new  = np.interp(t_new, t_old,
beta_p)     return np.concatenate([gamma_new, beta_new])   # ---- Example
orchestration: N=8, scale p=1 -> p=2 ---- if __name__ == '__main__':     from
qaoa_pennylane import make_qaoa_qnode      N = 8     circuit_p1, _ =
make_qaoa_qnode(N=N, p=1)     obj_p1 = lambda params, shots:
float(circuit_p1(params))   # replace shots for real HW      pso_p1 =
KitaevPSO(N=N, p=1, n_particles=20, max_iters=80, seed=0)     best_p1, E_p1 =
pso_p1.run(obj_p1)     print(f"p=1 best energy: {E_p1:.6f}")      # Warm-
start p=2 from p=1 result     init_p2 = interp_warmstart(best_p1)
circuit_p2, _ = make_qaoa_qnode(N=N, p=2)     obj_p2 = lambda params, shots:
float(circuit_p2(params))      pso_p2 = KitaevPSO(N=N, p=2, n_particles=25,
max_iters=100, seed=1)     pso_p2.rng = np.random.default_rng(1)     # Inject
warm-start particle as first particle     best_p2, E_p2 = pso_p2.run(obj_p2)
print(f"p=2 best energy: {E_p2:.6f}  (exact: -5.656854)")
```

Section 4: Noise Model Parameters and Mitigation

4.1 Hardware Noise Parameters

Representative calibration parameters for IBM superconducting devices relevant to this experiment. Eagle r3 (127-qubit heavy-hex) is the primary target for N = 8 and 10; Heron r2 (133-qubit) is required for N = 12 at p ≥ 3. Values below reflect daily calibration ranges observed in 2025–2026.

Parameter	Eagle r3 (conservative)	Eagle r3 (typical)	Heron r2 (optimistic)
T_1 (µs)	120	150	300
T_2 (µs)	80	100	200
1Q gate time (ns)	60	60	50
2Q gate time (ns)	400	350	200
Depol. 1Q	8×10^{-4}	5×10^{-4}	2×10^{-4}
Depol. 2Q (CX/ECR)	8×10^{-3}	6×10^{-3}	3×10^{-3}
Readout P(0|1)	2.5%	1.5%	0.8%
Readout P(1|0)	2.0%	1.0%	0.5%

Decoherence Risk: N=12, p=3 on Eagle r3

Estimated circuit time: $T_{circ} \approx 1800$ layers × 350 ns = 630 µs. This substantially exceeds T_1 = 150 µs. After optimization and DD: effective depth ≈ 1,200 gates, $T_{circ} \approx 420$ µs, $T_{circ}/T_1 \approx 2.8$. **Use Heron r2 for N=12 at p≥3.** On Heron r2: $T_{circ}/T_1 \approx 0.8$ — manageable with ZNE.

4.2 Qiskit Aer Noise Model Construction

```
from qiskit_aer.noise import (NoiseModel, depolarizing_error,
thermal_relaxation_error, ReadoutError)  def build_noise_model(T1=150e-6,
T2=100e-6, depol_1q=5e-4, depol_2q=6e-3,                      t_1q=60e-9,
t_2q=350e-9, p01=0.01, p10=0.015):     """     Construct a composite Aer
NoiseModel combining depolarizing noise,     thermal relaxation, and readout
errors.     T1, T2 in seconds; t_1q, t_2q = gate times in seconds.     p01 =
P(measure 0 | state 1); p10 = P(measure 1 | state 0).     """     nm =
NoiseModel()      # 1-qubit error: depolarizing + T1/T2 relaxation     err_1q
= (depolarizing_error(depol_1q,
1)               .compose(thermal_relaxation_error(T1, T2, t_1q)))      # 2-
qubit error: depolarizing + T1/T2 on each qubit     relax_2q =
(thermal_relaxation_error(T1, T2,
t_2q)                 .expand(thermal_relaxation_error(T1, T2, t_2q)))
err_2q   = depolarizing_error(depol_2q, 2).compose(relax_2q)      # Readout
assignment error (per-qubit)     ro_err = ReadoutError([[1 - p10, p10], [p01,
1 - p01]])      # Apply to all qubits
nm.add_all_qubit_quantum_error(err_1q,         ['rx', 'ry', 'rz', 'h', 's',
'sdg', 'sx', 'x'])     nm.add_all_qubit_quantum_error(err_2q, ['cx', 'cz',
'ecr', 'rxx', 'ryy'])     nm.add_all_qubit_readout_error(ro_err)     return
nm   # ---- Eagle r3 typical model ---- noise_eagle_typical =
build_noise_model()  # ---- Heron r2 optimistic model ---- noise_heron =
build_noise_model(T1=300e-6, T2=200e-6,
depol_1q=2e-4, depol_2q=3e-3,                                  t_1q=50e-9,
t_2q=200e-9,                                p01=0.008, p10=0.005)  # ----
Use with AerSimulator ---- from qiskit_aer import AerSimulator sim_noisy =
AerSimulator(noise_model=noise_eagle_typical, method='statevector')
```

4.3 Readout Error Calibration

Tensored readout calibration prepares each qubit independently in $|0\rangle$ and $|1\rangle$, measuring the 2N calibration circuits needed to build per-qubit 2×2 assignment matrices M_j, where $(M_j)_{kl} = P(\text{measure } k \mid \text{prepare } l)$. The full N-qubit assignment matrix is the tensor product $\bigotimes_j M_j$, which is then inverted (or pseudo-inverted) and applied to measured probability vectors to recover corrected probabilities.

N	Calibration circuits	Shots per circuit	Total calibration shots	Recalibration interval
8	16	8,192	131,072	Every 4 hours
10	20	8,192	163,840	Every 4 hours
12	24	8,192	196,608	Every 4 hours

4.4 Zero-Noise Extrapolation (ZNE)

Zero-Noise Extrapolation amplifies the circuit noise level by a set of integer scale factors $\lambda = \{1, 2, 3, \ldots\}$ using gate folding (replacing each gate G by $G \cdot G^\dagger \cdot G$ repeated k times for scale $2k+1$), then extrapolates the measured expectation values to the zero-noise limit $\lambda \to 0$ using Richardson extrapolation.

Richardson extrapolation weights for an (n)-point polynomial fit extrapolated to $\lambda = 0$:

$$
w
i
=
\prod
j \neq i
(0
-
\lambda
j
) / (
\lambda
i
-
\lambda
j
)
$$

For the standard 3-point scheme (λ = 1, 2, 3): w = (3, −3, 1) giving $E_{mitigated}$ = 3E(λ=1) − 3E(λ=2) + E(λ=3).

```
# zne.py — Richardson extrapolation and gate folding utilities import numpy as np  def richardson_extrapolate(scale_factors, expectation_values):     """ Exact Richardson extrapolation to zero noise.     scale_factors: list of noise amplification levels (e.g., [1, 2, 3])     expectation_values: corresponding measured <E> at each scale     Returns: float, extrapolated zero-noise expectation value.     """     x = np.array(scale_factors, dtype=float)     y = np.array(expectation_values, dtype=float)     n = len(x)     weights = np.array([         np.prod([(0.0 - x[j]) / (x[i] - x[j]) for j in range(n) if j != i])         for i in range(n)     ])     return float(np.dot(weights, y))   def fold_circuit(circuit, scale_factor: int): """     Gate folding for ZNE: replace each gate G with G (G_dag G)^k where scale_factor = 2k+1.  Requires scale_factor to be an odd integer >= 1. Returns a new QuantumCircuit with amplified noise.     """     assert scale_factor % 2 == 1 and scale_factor >= 1     k = (scale_factor - 1) // 2     if k == 0:         return circuit.copy()     folded = circuit.copy_empty_like()     for instruction in circuit.data:         folded.append(instruction)         for _ in range(k):         folded.append(instruction.operation.inverse(), instruction.qubits, instruction.clbits)         folded.append(instruction)     return folded   # ---- Example: 3-point ZNE ---- scales        = [1, 2, 3] noisy_energies = [-5.10, -4.75, -4.40]   # measured at each scale factor E_mitigated   = richardson_extrapolate(scales, noisy_energies) print(f"ZNE-mitigated energy: {E_mitigated:.4f}")   # -5.45 (approaching -5.66)  # For higher accuracy, use 5-point ZNE (scales [1,2,3,4,5]): scales5 = [1, 2, 3, 4, 5] # w = richardson_extrapolate(scales5, energies5)
```

4.5 Parity Postselection

```
def parity_postselect(counts: dict, target_parity: int = 0) -> tuple:     """ Filter measurement counts to retain only bitstrings of the target parity. target_parity=0: even parity (Kitaev ground state sector). target_parity=1: odd parity.     Returns (filtered_counts, acceptance_fraction).     Acceptance fraction diagnostics:       40-60%  : Normal (ideal ~50%, noise shifts this slightly)       20-40%  : Elevated noise or suboptimal mixer parameters       < 20%   : Critical — check parity-preserving mixer and transpilation     """     total   = sum(counts.values())     filtered = {         bs: cnt for bs, cnt in counts.items()         if bs.count('1') % 2 == target_parity     }     accepted = sum(filtered.values())     acceptance = accepted / total if total > 0 else 0.0     return filtered, acceptance   # ---- Usage example ---- raw_counts = {'00000000': 2100, '10100000': 890, '00001010': 750, '11111111': 230, '10000001': 350}  # mixed parity example even_counts, frac = parity_postselect(raw_counts, target_parity=0) print(f"Acceptance fraction : {frac:.2%}") print(f"Even-parity shots   : {sum(even_counts.values())}")
```

4.6 IBM Runtime Built-in Mitigation

```
from qiskit_ibm_runtime import Options  options = Options() # Resilience level 2: Zero-Noise Extrapolation with gate folding options.resilience_level = 2 # Optimization level 3: SABRE layout + routing + Peephole optimization options.optimization_level       = 3 # Dynamical decoupling with XY4 sequence (robust to pulse errors) options.dynamical_decoupling.enable        = True
```

```
options.dynamical_decoupling.sequence_type = 'XY4' # Execution settings
options.execution.shots = 8192  # Optional advanced ZNE settings
(resilience_level=2 exposes these): # options.resilience.noise_factors = [1,
2, 3]      # scale factors # options.resilience.extrapolator =
'LinearExtrapolator'  # or 'PolynomialExtrapolator'
```

Section 5: Hardware-Aware Transpilation

5.1 Target Devices

Device	Qubits	Topology	Native 2Q gate	Native 1Q gates	Recommended for
IBM Eagle r3 (ibm_brisbane)	127	Heavy-hex	CX (CNOT)	RZ, SX, X	N = 8, 10 at p ≤ 2
IBM Heron r2 (ibm_torino)	133	Heavy-hex	ECR	RZ, SX, X	N = 10–12, all p

The heavy-hex topology provides direct linear chains of length up to approximately 10–12 qubits without requiring any SWAP insertions for the Kitaev chain QAOA (which has only nearest-neighbor Pauli interactions). This makes linear chain embedding highly efficient on both target devices.

5.2 Qubit Selection Strategy

Selecting the best linear N-qubit chain from the device coupling graph is a critical preprocessing step, as 2Q gate error rates vary by a factor of 5–10 across qubit pairs on current devices. The selection procedure is:

1. Query `backend.properties()` for the latest calibration data (automatically refreshed at start of each session).

2. Extract the CX/ECR error rate for every available qubit pair in the coupling map.
3. Enumerate all connected linear paths of length N in the coupling graph using BFS/DFS (manageable for N ≤ 12).
4. Score each path by its total 2Q error rate: score = $\sum_{\text{bonds}}$ err($CX_{j,j+1}$).
5. Select the minimum-score path as the qubit layout.
6. For N = 12 on Eagle: one SWAP insertion may be unavoidable; include SWAP error in path scoring.

5.3 Transpilation Pipeline

```
from qiskit import transpile from qiskit.transpiler import CouplingMap from
qiskit_ibm_runtime import QiskitRuntimeService  def
select_best_linear_chain(backend, N: int):     """Find the N-qubit linear
chain with minimum total 2Q error."""     props = backend.properties()
coupling = backend.coupling_map     # Build error-weighted adjacency
cx_errors = {}     for gate_props in props.gates:         if gate_props.gate
in ('cx', 'ecr', 'cz') and len(gate_props.qubits) == 2:             q0, q1 =
gate_props.qubits             err = gate_props.parameters[0].value   #
error_rate parameter             cx_errors[(q0, q1)] = err
cx_errors[(q1, q0)] = err      # DFS over coupling graph to find best linear
path     best_path, best_score = None, np.inf     adj = {q: [p for (p, r) in
coupling] for q, p, r in coupling.get_edges()}      def dfs(path, visited,
score):         nonlocal best_path, best_score         if len(path) == N:
if score < best_score:                 best_score = score
best_path  = path.copy()             return         last = path[-1]
for neighbor in adj.get(last, []):             if neighbor not in visited:
e = cx_errors.get((last, neighbor), 0.01)
visited.add(neighbor)                 path.append(neighbor)
dfs(path, visited, score + e)                 path.pop()
visited.remove(neighbor)      for start_q in range(backend.num_qubits):
dfs([start_q], {start_q}, 0.0)      return best_path   # e.g., [42, 43, 44,
45, 46, 47, 48, 49] for N=8   # ---- Transpile with selected layout ---- def
transpile_kitaev(circuit, backend, qubit_layout):     N = circuit.num_qubits
return transpile(         circuit,         backend=backend,
initial_layout=qubit_layout,         optimization_level=3,        # SABRE
routing + Peephole optimization         seed_transpiler=42,          #
Reproducible layout         # basis_gates inferred from backend     )  #
Usage service = QiskitRuntimeService() backend =
service.backend('ibm_brisbane') layout  = select_best_linear_chain(backend,
N=8) print(f"Selected layout: {layout}") transpiled =
transpile_kitaev(bound_circuit, backend, layout) print(f"Depth   :
{transpiled.depth()}") print(f"CX count: {transpiled.count_ops().get('cx',
0)}")
```

5.4 Dynamical Decoupling

The XY4 pulse sequence (X–Y–X–Y) is applied during idle intervals to suppress low-frequency noise (1/f charge noise and flux noise dominant on transmons). It provides first-order decoupling from both X-type and Z-type noise, making it robust to control pulse imperfections compared to the simpler XX sequence.

```
from qiskit.transpiler.passes import ALAPScheduleAnalysis,
PadDynamicalDecoupling from qiskit.circuit.library import XGate, YGate from
qiskit.transpiler import PassManager, InstructionDurations  def
apply_xy4_dd(transpiled_circuit, backend):     """Apply XY4 dynamical
decoupling to idle windows."""     durations =
InstructionDurations.from_backend(backend)      dd_seq = [XGate(), YGate(),
XGate(), YGate()]   # XY4 sequence      pm =
PassManager([         ALAPScheduleAnalysis(durations=durations),
PadDynamicalDecoupling(             durations=durations,
dd_sequence=dd_seq,             pulse_alignment=16,   # align to 16dt grid
extra_slack_distribution='middle',          )     ])     return
pm.run(transpiled_circuit)  # Apply after transpilation dd_circuit =
apply_xy4_dd(transpiled, backend) print(f"DD circuit depth:
{dd_circuit.depth()}")
```

5.5 IBM Runtime Session

```
from qiskit_ibm_runtime import QiskitRuntimeService, Session, EstimatorV2 as
Estimator from qiskit.quantum_info import SparsePauliOp  def
run_kitaev_estimator(transpiled_circuit, H_sparse, options,
backend_name='ibm_brisbane'):     """     Execute QAOA circuit with
EstimatorV2 inside an IBM Runtime Session.     H_sparse: SparsePauliOp
representation of H_kitaev.     Returns measured energy expectation value.
"""     service = QiskitRuntimeService()     backend =
service.backend(backend_name)      with Session(backend=backend) as session:
estimator = Estimator(session=session, options=options)         # EstimatorV2
takes (circuit, observable) pairs         job    =
estimator.run([(transpiled_circuit, H_sparse)])         result = job.result()
energy = result[0].data.evs        # expectation value         std    =
result[0].data.stds       # standard error (if available)         print(f"
Energy: {energy:.6f} +/- {std:.6f}")     return energy  # Build SparsePauliOp
from PennyLane Hamiltonian (bridge utility) def
pennylane_H_to_sparse_pauli(H, N):     from qiskit.quantum_info import
SparsePauliOp     pauli_list = []     for coeff, op in zip(H.coeffs, H.ops):
pauli_str = ['I'] * N         for wire in op.wires:             name =
type(op).__name__.replace('Pauli','')             if hasattr(op, 'wires') and
len(op.wires) == 1:                 pauli_str[wire] = name
pauli_list.append((''.join(reversed(pauli_str)), coeff))     return
SparsePauliOp.from_list(pauli_list)
```

Section 6: Resource Estimates

6.1 Gate Count Estimates (Pre-Transpilation, Analytical)

Analytical formulas for gate counts before transpilation optimization. Each cost layer contributes $8(N-1)$ CX gates (4 Pauli rotation types × 2 CX each × $(N-1)$ bonds). Each mixer layer contributes $4(N-1)$ CX gates (2 types × 2 CX × $(N-1)$ bonds).

N	p	CX (cost)	CX (mixer)	CX total	SQ total (est.)	Depth (est.)	Parameters
8	1	56	28	84	620	380	2
8	2	112	56	168	1,240	760	4
10	2	144	72	216	1,580	980	4
10	3	216	108	324	2,370	1,470	6
12	2	176	88	264	1,920	1,200	4
12	3	264	132	396	2,880	1,800	6

After `optimization_level=3` transpilation, expect a 15–25% CX reduction from Peephole optimization (adjacent-gate cancellation and KAK decomposition of two-qubit blocks). For Heron r2 (ECR native gate), an additional 10% reduction is typical as ECR enables more compact Clifford synthesis.

6.2 Decoherence Budget Analysis

Circuit execution time governs the decoherence budget. The coherence factor for a circuit of duration T_{circ} is approximately $\exp(-T_{circ}/T_1)$ per qubit, assuming predominantly T_1-limited relaxation. The critical regime is $T_{circ}/T_1 > 1$, where error mitigation becomes insufficient.

N	p	Depth (post-opt.)	T_{circ} Eagle r3	T_{circ}/T_1 Eagle	T_{circ} Heron r2	T_{circ}/T_1 Heron	Recommendation
8	1	~290	~102 μs	0.68	~29 μs	0.10	Either

N	p	Depth (post-opt.)	T_{circ} Eagle r3	T_{circ}/T_1 Eagle	T_{circ} Heron r2	T_{circ}/T_1 Heron	Recommendation
8	2	~570	~200 µs	1.33	~57 µs	0.19	Heron preferred
10	2	~740	~259 µs	1.73	~74 µs	0.25	Heron preferred
10	3	~1,100	~385 µs	2.57	~110 µs	0.37	Heron required
12	2	~900	~315 µs	2.10	~90 µs	0.30	Heron required
12	3	~1,350	~473 µs	3.15	~135 µs	0.45	Heron required

Critical: N=12 Backend Requirement

For N=12 at any p, Eagle r3 yields $T_{circ}/T_1 > 2$ — severe decoherence dominates. Use Heron r2 (ibm_torino or equivalent) exclusively for N=12. For N=8–10 at p=1, Eagle r3 is acceptable. For p≥2, Heron r2 is strongly preferred.

6.3 Wall Time and Cost Estimates

Assumptions: 10-minute average queue wait per job batch; 4,096 shots per job submission; IBM Quantum pricing approximately $1.60 per runtime-second (Flex Plan, 2025–2026). All estimates are approximate and subject to device availability and plan conditions.

N	p	Total shots	Job batches	Exec time	Queue overhead	Total wall time	Est. cost (USD)
8	1	1,600,000	391	~45 min	10 min	~55 min	~$0.50

N	p	Total shots	Job batches	Exec time	Queue overhead	Total wall time	Est. cost (USD)
8	2	3,000,000	733	~85 min	10 min	~95 min	~$0.90
10	2	4,500,000	1,099	~125 min	10 min	~135 min	~$1.30
10	3	7,000,000	1,709	~195 min	10 min	~205 min	~$2.00
12	2	6,000,000	1,465	~170 min	10 min	~180 min	~$1.75
12	3	9,000,000	2,198	~250 min	10 min	~260 min	~$2.55

Note: Cost estimates are approximate. Actual pricing depends on IBM Quantum plan tier, specific device, current queue conditions, and whether ZNE scale factors multiply the shot budget. ZNE with 3 scale factors triples the hardware shot count, proportionally increasing both time and cost estimates above.

Section 7: Validation Suite

7.1 Metrics Overview

Metric	Formula	Pass threshold (sim.)	Pass threshold (HW)	Notes
Energy relative error	$\|E_{QAOA} - E_{exact}\| / \|E_{exact}\|$	< 2%	< 8%	Primary convergence metric
State fidelity	$\|\langle\psi_{exact}\|\psi_{QAOA}\rangle\|^2$	> 0.90	N/A (statevector only)	Statevector simulation only
Parity expectation	$\langle\prod_j Z_j\rangle$	> 0.99	> 0.85	Must be +1 for even sector
String order	$\langle Z_0 X_1 \ldots X_{N-2}$	Non-zero in	Non-zero (ZNE	Topological

Metric	Formula	Pass threshold (sim.)	Pass threshold (HW)	Notes
parameter	$Z_{N-1}\rangle$	topo. phase	corrected)	order indicator
Majorana correlator	$\langle X_0 Z_1 ... Z_{N-2} Y_{N-1}\rangle$	Non-zero	Non-zero (ZNE corrected)	Edge mode detection
Boundary occupation	$\langle n_0\rangle$, $\langle n_{N-1}\rangle$	~0.5	~0.5 ± 0.15	Delocalized Majorana signature

7.2 Energy Validation Code

```
import numpy as np from numpy.linalg import eigh from itertools import
product  def build_qubit_hamiltonian(N: int, mu: float = 0., w: float = 1.,
delta: float = 1.):     """     Construct the full 2^N x 2^N qubit
Hamiltonian matrix for exact diagonalization.     Uses direct Pauli matrix
construction (efficient for N <= 14).     """     I = np.eye(2,
dtype=complex)     X = np.array([[0,1],[1,0]], dtype=complex)     Y =
np.array([[0,-1j],[1j,0]], dtype=complex)     Z = np.array([[1,0],[0,-1]],
dtype=complex)      def kron_op(ops: list) -> np.ndarray:         """Tensor
product of a list of 2x2 matrices."""         result = ops[0]         for op
in ops[1:]:             result = np.kron(result, op)         return result
def pauli_string(pauli_dict: dict, N: int) -> np.ndarray:         """Build N-
qubit Pauli string. pauli_dict: {qubit_idx: 'X'/'Y'/'Z'}."""
pauli_map = {'I': I, 'X': X, 'Y': Y, 'Z': Z}         return
kron_op([pauli_map[pauli_dict.get(j, 'I')] for j in range(N)])      dim = 2
** N     H   = np.zeros((dim, dim), dtype=complex)      # Z terms (chemical
potential)     for j in range(N):         H += (mu / 2.) * pauli_string({j:
'Z'}, N)      # Hopping (XX + YY) and pairing (XY - YX) terms     for j in
range(N - 1):         H += (-w    / 2.) * pauli_string({j: 'X', j+1: 'X'}, N)
H += (-w    / 2.) * pauli_string({j: 'Y', j+1: 'Y'}, N)         H += (-delta/
2.) * pauli_string({j: 'X', j+1: 'Y'}, N)         H += ( delta/ 2.) *
pauli_string({j: 'Y', j+1: 'X'}, N)      return H   def exact_ground_state(N:
int, mu: float = 0., w: float = 1., delta: float = 1.):     """Return (E0,
psi0) from dense diagonalization. psi0 is in even parity sector."""     H =
build_qubit_hamiltonian(N, mu, w, delta)     evals, evecs = eigh(H)
return float(evals[0].real), evecs[:, 0]   def validate_energy(E_qaoa: float,
N: int, mu: float = 0., w: float = 1., delta: float = 1.):     E_exact, _ =
exact_ground_state(N, mu, w, delta)     rel_err = abs(E_qaoa - E_exact) /
abs(E_exact)     status = "PASS" if rel_err < 0.02 else ("WARN" if rel_err <
0.08 else "FAIL")     print(f"[{status}] E_QAOA={E_qaoa:.6f},
E_exact={E_exact:.6f}, RelErr={rel_err:.4%}")     return rel_err, status   #
---- Verification (run before hardware experiments) ---- if __name__ ==
'__main__':     for N in [8, 10, 12]:         E0, _ = exact_ground_state(N)
print(f"N={N:2d}: E0 = {E0:.6f}  (expected {-N*np.sqrt(2)/2:.6f})")     #
Expected output:     # N= 8: E0 = -5.656854  (expected -5.656854)     # N=10:
E0 = -7.071068  (expected -7.071068)     # N=12: E0 = -8.485281  (expected -
8.485281)
```

7.3 Parity and Edge Observable Code

```
import numpy as np  def parity_expectation(statevector: np.ndarray, N: int) -> float:     """     Compute <prod_j Z_j> from statevector.     Returns +1.0 for even parity ground state, -1.0 for odd.     """     # Z^{tensor N} eigenvalue for basis state |k> = (-1)^(popcount(k))     z_evals = np.array([(-1) ** bin(k).count('1') for k in range(2 ** N)])     probs   = np.abs(statevector) ** 2     return float(probs @ z_evals)   def boundary_occupation(statevector: np.ndarray, N: int) -> tuple:     """     Compute <n_0> and <n_{N-1}> = <(I - Z_j)/2> at boundary sites.     Signature of delocalized Majorana: both should be ~0.5.     """     probs = np.abs(statevector) ** 2     # <n_j> = sum over basis states where qubit j is |1>     n0   = sum(probs[k] for k in range(2**N) if (k >> (N-1)) & 1)   # qubit 0 (MSB convention)     nN1  = sum(probs[k] for k in range(2**N) if k & 1)               # qubit N-1 (LSB)     return float(n0), float(nN1)   def fidelity(sv_qaoa: np.ndarray, N: int, mu: float = 0., w: float = 1., delta: float = 1.) -> float:     """Compute |<psi_exact|psi_qaoa>|^2."""     _, psi_exact = exact_ground_state(N, mu, w, delta)     overlap = np.dot(psi_exact.conj(), sv_qaoa)     return float(abs(overlap) ** 2)   def string_order_parameter(statevector: np.ndarray, N: int) -> float:     """     Non-local string order: <Z_0 X_1 X_2 ... X_{N-2} Z_{N-1}>     Non-zero value signals topological order (Kitaev phase |mu| < 2).     """     from numpy.linalg import norm     I = np.eye(2, dtype=complex)     X = np.array([[0,1],[1,0]], dtype=complex)     Z = np.array([[1,0],[0,-1]], dtype=complex)     op_list = [Z] + [X] * (N - 2) + [Z]     O = op_list[0]     for m in op_list[1:]:         O = np.kron(O, m)     psi = statevector     return float((psi.conj() @ O @ psi).real)   # ---- Full validation report ---- def full_validation_report(sv_qaoa: np.ndarray, E_qaoa: float, N: int):     print(f"\n{'='*60}")     print(f"VALIDATION REPORT: N={N}")     print(f"{'='*60}")     rel_err, status = validate_energy(E_qaoa, N)     F = fidelity(sv_qaoa, N)     P      = parity_expectation(sv_qaoa, N)     SOP = string_order_parameter(sv_qaoa, N)     n0, nN = boundary_occupation(sv_qaoa, N)     print(f"  Fidelity          : {F:.4f}  ({'PASS' if F>0.90 else 'FAIL'})")     print(f"  Parity <P>        : {P:.4f}  ({'PASS' if P>0.99 else 'FAIL'})")     print(f"  String order      : {SOP:.4f}  (non-zero = topological)")     print(f"  Boundary occ. n_0 : {n0:.4f}, n_{{N-1}}: {nN:.4f}  (target ~0.5)")     print(f"{'='*60}\n")
```

7.4 Phase Diagram Analysis

```
import numpy as np import matplotlib matplotlib.use('Agg') import matplotlib.pyplot as plt  def compute_phase_diagram(N: int, n_points: int = 25, w: float = 1., delta: float = 1.):     """     Sweep mu from -3 to +3 and compute exact ground-state energies     and string order parameters. Phase boundary at mu = +/-2 for w=1.     """     mu_values = np.linspace(-3., 3., n_points)     energies  = []     sops      = []     for mu in mu_values:         E0, psi0 = exact_ground_state(N, mu, w, delta)         energies.append(E0)         sops.append(string_order_parameter(psi0, N))     return mu_values, np.array(energies), np.array(sops)  mu_vals, E_exact, SOP = compute_phase_diagram(N=8) # Topological phase boundary: |mu| = 2 # String order: non-zero for |mu| < 2, vanishes for |mu| > 2 # Energy minimum: deepest in the center of the topological phase (mu=0) print("Phase boundary verification:") for i, mu in enumerate(mu_vals):     phase = "TOPO" if abs(mu) < 2. else "TRIVIAL"     print(f"  mu={mu:+.2f}: E={E_exact[i]:.4f}, SOP={SOP[i]:+.4f} [{phase}]")
```

7.5 Convergence Plots and Analysis

The following plots should be generated after each experimental phase and stored in the `results/` directory:

7. **PSO convergence plot:** Best energy vs. PSO iteration number, with the exact E_0 reference line and the adaptive shot schedule overlaid as a step function on a secondary y-axis.
8. **Adaptive shot schedule:** Shots per evaluation vs. iteration (Phase 1: 256, Phase 2: 1,024, Phase 3: 4,096).
9. **Phase diagram:** Ground-state energy and string order parameter vs. mu, with shading for |mu| < 2 (topological) and |mu| > 2 (trivial) regions. Overlay QAOA energies at the converged parameters.
10. **Fidelity scaling:** Fidelity F vs. system size N for p = 1, 2, 3 from noiseless statevector simulation. Identify minimum p required for F > 0.90 as a function of N.
11. **ZNE mitigation comparison:** Raw vs. ZNE-mitigated energy for each hardware run, with exact E_0 reference. Report improvement factor = (raw error) / (mitigated error).

Section 8: Runbook — Milestones and Fallback Strategies

8.1 Project Timeline (6 Weeks)

Week	Phase	Key Activities	Deliverables
Week 1	Phase 0: Setup	Environment installation, exact diagonalization, circuit building tests	All checkpoints in §8.2 passing; baseline energy table verified

Week	Phase	Key Activities	Deliverables
Week 2	Phase 1a: Noiseless sim.	N=8 PSO convergence, fidelity measurement, phase diagram (noiseless)	N=8, p=2: F>0.90, RelErr<2%; PSO convergence plot
Week 3	Phase 1b: Noisy sim.	Noise model construction, ZNE testing, parity postselection validation	ZNE reduces RelErr from >10% to <5% on noisy simulator
Week 4	Phase 2a: HW N=8,10	Hardware execution N=8,10; transpilation; readout calibration	N=8 HW RelErr < 8%; N=10 HW RelErr < 12%
Week 5	Phase 2b: HW N=12	N=12 on Heron r2; full mitigation pipeline; parity postselection	N=12 HW energy within 15% of exact; parity acceptance > 40%
Week 6	Phase 3: Analysis	Phase diagram scan, fidelity scaling, ZNE comparison, report	All plots generated; final results table; report draft

8.2 Phase 0: Environment Setup (Week 1)

Installation:

```
pip install qiskit>=2.3 qiskit-aer>=0.14 qiskit-ibm-runtime>=0.25 pip install pennylane>=0.45 pennylane-qiskit scipy matplotlib networkx pip install numpy>=1.26
```

Validation Checkpoints (all must pass before hardware execution):

- ☐ N=8 exact $E_0 = -5.656854 \pm 1\times10^{-5}$
- ☐ N=10 exact $E_0 = -7.071068 \pm 1\times10^{-5}$
- ☐ N=12 exact $E_0 = -8.485281 \pm 1\times10^{-5}$
- ☐ Parity $\langle P \rangle = +1.000$ for exact ground states at all N
- ☐ QAOA circuit builds without error for all (N, p) in {8,10,12} × {1,2,3,4}
- ☐ PennyLane QNode returns finite energy at random initial parameters

- ☐ PSO mock objective (random function) converges without error in < 50 iterations
- ☐ Noise model constructs without error; Aer simulation runs at N=8
- ☐ ZNE Richardson extrapolation unit test: 3-point test case matches analytic weights
- ☐ IBM Quantum account authenticated; at least one backend accessible

8.3 Phase 1: Simulator Milestones

Milestone 1.1 — Noiseless N=8, p=2

Target: Fidelity F > 0.90, energy relative error < 2%.

Method: Run PSO (25 particles, 100 iterations) with PennyLane statevector device (shots=None).

Fallback: If F < 0.90 at p=2, increase to p=3 using INTERP warm-start from p=2 solution. Verify parity expectation > 0.99 before proceeding.

Milestone 1.2 — Shot-Noise Robustness N=8

Target: Mean energy within 3% of exact over 10 independent PSO trials at 4,096 shots.

Method: Replace statevector device with `shots=4096`; run 10 seeds; report mean and standard deviation of final energies.

Fallback: If mean error > 3%: increase Phase 3 shots to 8,192; apply ZNE on the shot-based simulator using Pauli twirling to verify extrapolation correctness.

Milestone 1.3 — Scale N=10, 12 (Noiseless)

Targets: N=10 fidelity ≥ 0.85 at p=3; N=12 fidelity ≥ 0.80 at p=3.

Method: Use INTERP warm-start between system sizes (transfer N=8 optimal params to N=10 via interpolation). Scale PSO hyperparameters per Section 3.3 table.

Milestone 1.4 — Noisy Simulator (Eagle r3 Noise Model)

Target: ZNE recovery brings relative energy error from >10% (raw noisy) to <5% (mitigated). Demonstrate parity acceptance fraction > 40%.

Method: Use Aer with Eagle r3 typical noise model (Section 4.2). Apply 3-point ZNE ($\lambda = 1, 2, 3$) with gate folding. Verify parity postselection acceptance.

8.4 Phase 2: Hardware Milestones

Milestone 2.1 — N=8 Hardware Baseline

Backend: ibm_brisbane (Eagle r3) or ibm_torino (Heron r2).

Settings: 8,192 shots; resilience_level=2; XY4 DD; optimization_level=3.

Target: Hardware energy within 8% of exact (−5.656854).

Pre-run checks: Verify qubit T_1, T_2 from calibration data; confirm selected linear chain has all CX error rates below 1%.

Milestone 2.2 — Readout Calibration

Run 2N calibration circuits at the start of each hardware session. Store per-qubit assignment matrices in `calibration/` directory with timestamp. Re-calibrate if session exceeds 4 hours (thermal drift changes readout fidelity significantly on timescales of hours).

Milestone 2.3 — N=10 Hardware

Strategy: Use PSO-converged parameters from simulator as fixed evaluation point (no hardware re-optimization). Perform single hardware evaluation at converged γ^*, β^* with 8,192 shots and ZNE.

Fallback: If Eagle queue wait exceeds 2 hours, switch to Heron r2 backend. Note device change in results metadata.

Milestone 2.4 — N=12 Hardware (Heron r2 Required)

Backend: ibm_torino (Heron r2) — mandatory, not Eagle r3.

Expected performance: Fidelity 0.45–0.65 even with full mitigation pipeline (significant noise at this depth/size combination is expected and acceptable).

Mitigation stack: Readout calibration + ZNE (scales [1,2,3]) + parity postselection + XY4 DD.

8.5 Phase 3: Analysis (Week 6)

Milestone 3.1 — Phase Diagram Scan

25-point mu sweep on the noiseless simulator using the converged parameters from mu=0 optimization (no re-optimization at each mu point, to test parameter transferability). Overlay exact and QAOA energies; shade topological region. Identify the sharpening of the phase transition with increasing N.

Milestone 3.2 — Fidelity Scaling Analysis

Fit F(N) scaling curve for each p = 1, 2, 3 from noiseless statevector simulation. Report minimum p required to achieve F > 0.90 as a function of N. This informs the resource requirements for larger-scale experiments.

Milestone 3.3 — ZNE Effectiveness Quantification

For each hardware run, report: raw energy, ZNE-mitigated energy, exact energy, and the ZNE improvement factor. Compare 3-point Richardson extrapolation vs. linear fit. Determine if 5-point ZNE provides additional improvement for N=12.

8.6 Fallback Decision Tree

Symptom	Diagnosis	Corrective Action
Energy error > 15% after full mitigation	Severe hardware noise or wrong parity sector	1. Check parity acceptance; if <20%, fix mixer. 2. Increase ZNE scales to [1,3,5]. 3. Switch to Heron r2. 4. Reduce p by 1.
Parity acceptance < 20%	Mixer not parity-preserving; transpilation error	Verify RXX+RYY decomposition; check basis gate compilation; inspect transpiled circuit manually.
PSO stagnation (>20 iterations no improvement)	Barren plateau or insufficient exploration	Automatic restart with scattered particles (built-in); INTERP warm-start

Symptom	Diagnosis	Corrective Action
		from p−1; try Adam-FIPSO hybrid (arXiv:2506.06790).
$T_{circ} >> T_2$ (decoherence dominant)	Circuit too deep for current hardware	Switch to Heron r2 (T_1=300 μs); reduce p by 1; consider error detection encoding (two-qubit repetition code for parity).
String order parameter = 0 on hardware	Wrong phase or excessive noise on long string operator	Apply ZNE specifically to string order measurement; verify mu=0 initialization; check operator decomposition.
Readout acceptance > 80% (too high)	Readout bias toward $\|0\rangle$ dominating; noise mixing states	Re-calibrate readout; check assignment matrix condition number; verify calibration timestamp is < 4 hours old.

8.7 Complete Resource Summary

N	p	Target F (sim.)	Target F (HW)	CX total	Circuit depth	Total shots	Wall time	Backend
8	2	≥ 0.95	≥ 0.60	168	760	3.0M	~95 min	Eagle / Heron
10	3	≥ 0.85	≥ 0.50	324	1,470	7.0M	~205 min	Eagle / Heron
12	3	≥ 0.70	≥ 0.45	396	1,800	9.0M	~260 min	**Heron r2 only**

Note on Hardware Fidelity Targets

Hardware fidelity cannot be measured directly (no statevector access). The hardware "fidelity" target above is inferred from the energy error and ZNE mitigation quality. For hardware, the primary success criterion is energy relative

error < 8% (N=8), < 12% (N=10), < 15% (N=12) after ZNE and parity postselection.

Appendix A: File Structure

```
kitaev_qaoa_pack/ | |-- main_experiment.py            # Orchestrator; CLI entry point |   # Usage: python main_experiment.py --N 8 --p 2 --backend simulator |   # Usage: python main_experiment.py --scale         # run N=8->10->12 |   # Usage: python main_experiment.py --N 8 --backend eagle --shots 8192 | |-- hamiltonians/ |   |-- kitaev_mapping.py         # JW mapping, dense matrix, exact diagonalization |   |-- pauli_utils.py            # Pauli string utilities, SparsePauliOp converters | |-- circuits/ |   |-- qaoa_qiskit.py            # Qiskit QAOA circuit templates, resource estimation |   |-- qaoa_pennylane.py         # PennyLane QNode factory, IsingXX/YY cost+mixer |   |-- initial_states.py         # Half-filling state preparation utilities | |-- optimizer/ |   |-- pso_optimizer.py          # Full PSO with adaptive shots, restarts, INTERP |   |-- callbacks.py # Logging, convergence tracking, early stopping | |-- noise/ |   |-- noise_mitigation.py       # Noise models, readout calibration, ZNE, postselect |   |-- readout_calibration.py    # Tensored calibration protocol, matrix inversion |   |-- zne.py                    # Gate folding, Richardson extrapolation | |-- transpilation/ |   |-- hardware_transpilation.py # Transpile pipeline, qubit selection, DD |   |-- ibm_runtime_session.py    # IBM Runtime session template, EstimatorV2 wrapper | |-- validation/ |   |-- validation_suite.py       # Energy, fidelity, edge observables, phase diagram |   |-- plotting.py               # All matplotlib figure generation | |-- runbook/ |   |-- runbook.md                # This runbook in Markdown format |   |-- checklist.md              # All milestone checkboxes (auto-updated by CI) | |-- results/                      # Auto-created; stores all experimental outputs |   |-- simulator/ |   |-- hardware/ |   |-- calibration/ | |-- requirements.txt |-- README.md
```

Appendix B: Quick Start (5 Minutes)

```
# 1. Clone and install dependencies cd kitaev_qaoa_pack pip install -r requirements.txt  # 2. Validate exact diagonalization (no quantum hardware needed) python hamiltonians/kitaev_mapping.py # Expected output: #   N= 8: E0 = -5.656854    <P> = +1.0000 #   N=10: E0 = -7.071068    <P> = +1.0000 #   N=12: E0 = -8.485281    <P> = +1.0000  # 3. Run noiseless simulator experiment for N=8, p=2 python main_experiment.py --N 8 --p 2 --backend simulator # Expected: RelErr < 2%, Fidelity > 0.90 after PSO convergence  # 4. Run noisy simulator with Eagle r3 noise model python main_experiment.py --N 8 --p 2 --backend noisy_simulator --zne  # 5. Run full scaling experiment N=8 -> N=10 -> N=12 (simulator) python main_experiment.py --scale --backend
```

```
simulator  # 6. Hardware run (requires IBM Quantum account and active
session) #    Authenticate first: python -c "from qiskit_ibm_runtime import
QiskitRuntimeService; \ #
QiskitRuntimeService.save_account(channel='ibm_quantum', token='YOUR_TOKEN')"
python main_experiment.py --N 8 --p 2 --backend eagle --shots 8192  # 7. N=12
hardware run (Heron r2 only) python main_experiment.py --N 12 --p 3 --backend
heron --shots 8192 --zne  # 8. Generate all validation plots from saved
results python validation/plotting.py --results results/simulator/N8_p2/
```

Appendix C: Key References

Reference	Topic	Relevance
Kitaev, A. Yu. (2001). "Unpaired Majorana fermions in quantum wires." *Physics-Uspekhi* 44, 131–136.	Original Kitaev chain model	Primary Hamiltonian definition; topological phase analysis
Farhi, E., Goldstone, J., & Gutmann, S. (2014). "A Quantum Approximate Optimization Algorithm." arXiv:1411.4028.	QAOA algorithm	Original QAOA formulation; ansatz structure; motivation
Bhat, Wang, & Parampalli (2025). "Benchmarking Swarm Optimization for QAOA." arXiv:2506.06790v3.	PSO for QAOA	PSO hyperparameter recommendations; shot-noise benchmarking; Adam-FIPSO hybrid
Mohammadipour & Li (2025). "Direct Analysis of ZNE." *Quantum* 9, 1909.	Zero-Noise Extrapolation	Richardson extrapolation accuracy; gate folding protocols
Zhou, L., et al. (2020). "Quantum Approximate Optimization Algorithm: Performance, Mechanism, and Implementation." *Physical Review X* 10, 021067. arXiv:1812.01041.	QAOA warm-starting (INTERP)	INTERP parameter transfer strategy; landscape analysis
Kandala, A., et al. (2017). "Hardware-efficient variational quantum eigensolver for small molecules." *Nature* 549, 242–246.	VQE on superconducting hardware	Hardware noise mitigation; readout calibration protocols

Reference	Topic	Relevance
PennyLane Documentation, Xanadu (2025). pennylane.ai, v0.45.	PennyLane framework	IsingXX, IsingYY, PauliRot gates; parameter-shift rules; QNode design
Qiskit Documentation, IBM (2025). qiskit.org, v2.3+.	Qiskit framework	Circuit transpilation; EstimatorV2; Aer noise simulation; IBM Runtime
IBM Quantum Documentation (2025). "IBM Heron r2 Architecture." quantum.ibm.com.	Hardware specifications	Device connectivity; native gates; gate fidelity specifications

Document prepared by Robert on Tuesday, 26 May 2026 at 09:30 EDT. Lacey, NJ, United States. This document is a research-grade technical specification intended for use by qualified quantum computing researchers. All code is provided as a starting template; verify correctness against your specific Qiskit and PennyLane versions before hardware execution. IBM device specifications and pricing are subject to change; consult quantum.ibm.com for current values.